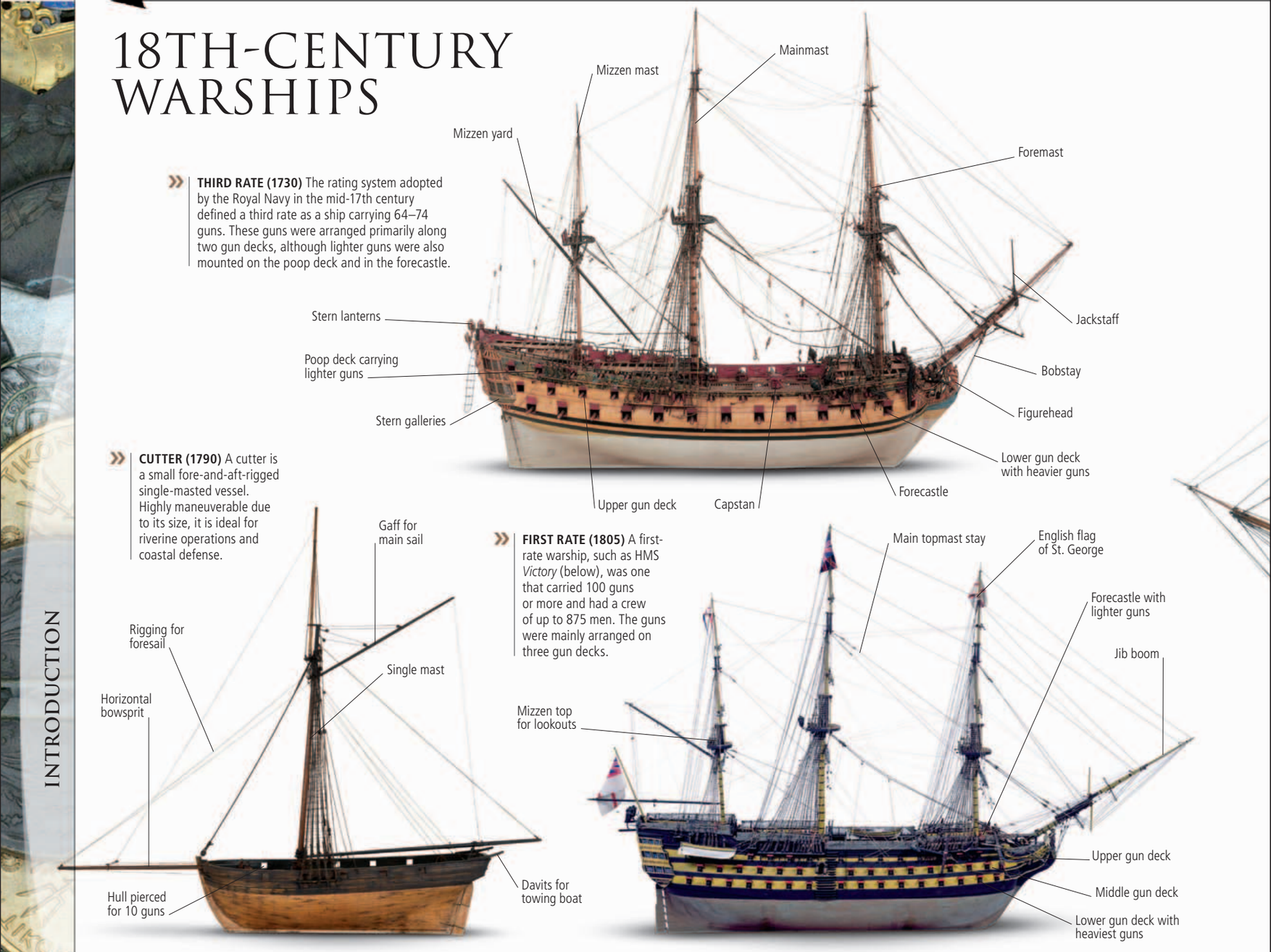
18TH-CENTURY WARSHIPS



THIRD RATE (1730) The rating system adopted by the Royal Navy in the mid-17th century defined a third rate as a ship carrying 64–74 guns. These guns were arranged primarily along two gun decks, although lighter guns were also mounted on the poop deck and in the forecastle.

CUTTER (1790) A cutter is a small fore-and-aft-rigged single-masted vessel. Highly maneuverable due to its size, it is ideal for riverine operations and coastal defense.

FIRST RATE (1805) A first-rate warship, such as HMS *Victory* (below), was one that carried 100 guns or more and had a crew of up to 875 men. The guns were mainly arranged on three gun decks.

By the 18th century oar power as a means of propulsion in battle had finally fallen by the wayside in most European navies, although galleys still served effectively in the shallow waters of the Baltic. The three-masted, multiple-gun-deck sailing ship, fighting in line of battle, remained unchallenged as the primary warship.

THE RATING SYSTEM

The design of warships became relatively standardized. They were categorized according to a rating system developed by England's Royal Navy in the mid-17th century, which referred to the number of guns a ship carried. By the mid-18th century a first-rate was a three-deck ship with 100 or more guns, while a second-rater had 90–98 guns also on three decks. The standard ship of the line was a two-decker third-rate, typically around 170 ft (52 m) long and mounting from 64 to 74 guns. Fourth-rate ships, with around 50 guns, were sometimes admitted as ships of the line. The naval ships not considered fit for the line of battle included frigates, sloops-of-war, schooners, brigs, and mortar vessels (variously fifth, sixth and unrated ships), which all nevertheless performed vital support functions. Frigates became especially important in the course of the 18th century. Fast and well-armed for their size, they served to great effect as scouts and commerce raiders. The frigates of the fledgling US Navy, such as *Constitution* launched at the end of the century, could carry as many as 60 guns and were a match for any fourth-rate ship of the line.

NEW TECHNOLOGIES

The warships of the 18th century benefited from some crucial technological advances. Firstly, ships' wheels had been introduced to replace the previous tiller system of steering. The pulley mechanism allied to the wheel allowed much greater and swifter movement of the rudder, and hence improved maneuverability. During the late 18th century copper sheathing of hulls was introduced. By protecting the hull, the copper plate allowed warships to spend longer at sea before requiring dry-dock maintenance and repairs. The introduction of the carronade in the 1770s provided a short-range but powerful gun light enough to be mounted on the upper deck without destabilizing the ship—the heaviest guns, such as the 32-lb Blomfields, had to be carried close to the waterline to maintain stability.